

MATH 123 LECTURE NOTES

WEEK 1

Syllabus Preliminaries, Limits and Continuity, Differentiation, More Applications of Differentiation, Integration, Techniques of Integration, Sequences, Series and Power Series.

PRELIMINARIES

Real Numbers and the Real Line

→ Real numbers are numbers that can be expressed as decimals, for

example:

(The sequence of decimal digits goes on forever)

$$5 = 5.0000 \dots$$

$$-\frac{3}{4} = -0.75000 \dots$$

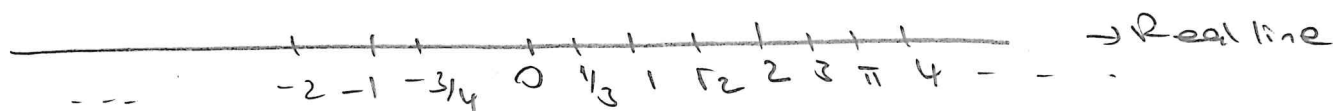
$$\frac{1}{3} = 0.3333 \dots$$

$$\sqrt{2} = 1.4142 \dots$$

$$\pi = 3.14159 \dots$$

Obvious patterns: we know what all the subsequent digits are

no obvious patterns.



The properties of the real number system fall into three categories:

- Algebraic Properties: Addition, Subtraction, Division, - - -
- Order Properties: The order of real numbers on the real line
- Completeness Property: If A is any set of real numbers having at least one number in it, and if there exists a real number with property $x \leq y$ for every x in A . Then, there exists a smallest number y in the same property. Roughly speaking, (1) there is no gap on real line. Every point on the real line is a real number.

ORDER IN REALS



$$a < b \quad \text{or} \quad b > a$$

$$a \leq b: \text{ either } a < b \quad \text{or} \quad a = b$$

$$b \geq a: \text{ either } b > a \quad \text{or} \quad b = a$$

Rules for inequalities

If a, b and c are real numbers then:

$$(1) \quad a < b \Rightarrow a + c < b + c$$

$$(2) \quad a < b \Rightarrow a - c < b - c$$

$$(3) \quad a < b \text{ and } c > 0 \Rightarrow a \cdot c < b \cdot c$$

$$(4) \quad a < b \text{ and } c < 0 \Rightarrow a \cdot c > b \cdot c; \text{ in particular } -a > -b$$

$$(5) \quad a > 0 \Rightarrow \frac{1}{a} > 0$$

$$(6) \quad 0 < a < b \Rightarrow \frac{1}{b} < \frac{1}{a}$$

Rules 1-4 and 6 (for $a > 0$) also holds if $<$ and $>$ are replaced by $\leq$ and $\geq$.

The set of real numbers has some special subsets:

(i) the natural numbers or positive integers, namely, the numbers $1, 2, 3, 4, \dots$

(ii) the integers namely the numbers $0, \pm 1, \pm 2, \pm 3, \dots$

(iii) the rational numbers, that is, numbers that can be expressed in the form of a fraction m/n where m and n are integers and $n \neq 0$

The rational numbers are precisely those real numbers with decimal expansions that are either:

(a) terminating, that is, ending with an infinite string of zeros,
severely
class 5th

for example $3/4 = 0.75000 \dots$

(b) repeating, that is ending with a string of digits that repeats over and over for example $23/11 = 2.090909 \dots = 2.\overline{09}$ (The bar indicates the pattern of repeating digits).

Real numbers that are not rational are called irrational numbers.

Ex 7 Show that each of the numbers (a) $1.323232 \dots = 1.\overline{32}$ and (b) $0.3405405405 \dots = 0.3\overline{405}$ is a rational number by expressing it as a quotient of integers.

(a) $x = 1.323232 \dots$ $10x - 1 = 0.323232 \dots$

$$100x = 132.3232 \dots = 132 + 0.323232 \dots = 132 + x - 1$$

$$99x = 131 \Rightarrow x = 131/99$$

(b) $y = 0.3405405405 \dots$ $10y = 3.405405405 \dots$

$$10y - 3 = 0.405405405 \dots$$

$$1000y = 3405.405405405 \dots = 3405 + 10y - 3$$

$$9990y = 3402 \text{ and } y = 3402/9990 = \frac{63}{185}$$

Intervals

A subset of the real line is called an interval if it contains at least two numbers and also contains ~~contains~~ all real numbers between any two of its elements. (1.1)

If a and b real numbers and $a < b$, we often refer

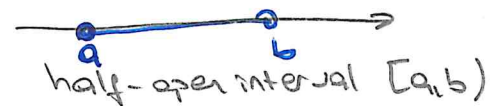
(i) the open interval from a to b , denoted by (a, b) consisting of all real numbers satisfying $a < x < b$



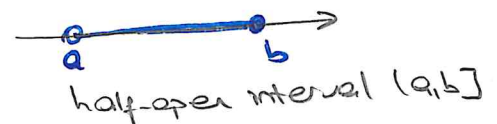
(ii) the closed interval from a to b , denoted by $[a, b]$ consisting of all real numbers x satisfying $a \leq x \leq b$



(iii) the half-open interval $[a, b)$ consisting of all real numbers x satisfying the inequalities $a \leq x < b$



(iv) the half-open interval $(a, b]$ consisting of all real numbers x satisfying the inequalities $a < x \leq b$



All have the finite length $b - a$.

Infinite intervals



the interval $[a, \infty)$



the interval $(-\infty, a]$



the real line $(-\infty, \infty)$

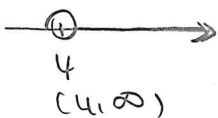
Example 1 Solve the following inequalities and express the solution sets in terms of intervals and graph them

(a) $2x-1 > x+3$

(b) $-\frac{x}{3} > 2x-1$

(c) $\frac{2}{x-1} \geq 5$

(a) $x > 4$



$(4, \infty)$

(b) $-x \geq 6x-3$
 $3 \geq 7x \quad x \leq \frac{3}{7}$
 $x \in (-\infty, \frac{3}{7}]$



(c) $\frac{2}{x-1} \geq 5$
 $\frac{2-5(x-1)}{x-1} \geq 0$
 $\frac{7-5x}{x-1} \geq 0$

$\frac{7-5x}{x-1} \geq 0 \Rightarrow$ roots $x = 7/5$ undefined at $x = 1$

x	1	7/5	
7-5x	+	+	0 -
x-1	-	0	+
$\frac{7-5x}{x-1}$	-	undef	+

The solution is $(1, 7/5]$



2 Solve the systems of inequalities

(a) $3 \leq 2x+1 \leq 5$

(b) $3x-1 < 5x+3 \leq 2x+15$

(a) $2 \leq 2x \leq 4 \quad 1 \leq x \leq 2 \quad x \in [1, 2]$

(b) $3x-1 < 5x+3 \Rightarrow -3-1 < 2x \quad \text{or} \quad 6x+3 \leq 2x+15$
 $\underline{-2 < x} \quad \underline{3x \leq 12} \quad \underline{x \leq 4}$
 $[-2, 4]$

3 Quadratic inequalities

(a) $x^2-5x+6 < 0$

(b) $2x^2+174x$

(a) $x^2-5x+6 = (x-3)(x-2) < 0$ it is negative if and only if one of them is negative.

(i) $x-3 < 0 \quad x-2 > 0$
 $x < 3 \quad x > 2 \quad \left. \vphantom{\begin{matrix} x < 3 \\ x > 2 \end{matrix}} \right\} 2 < x < 3$
 $(2, 3)$

(ii) $x-2 < 0 \quad x-3 > 0$
 $x < 2 \quad x > 3 \quad \text{No solution.}$

$$\textcircled{5} 2x^2 + 1 > 4x$$

$$2x^2 - 4x + 1 > 0$$

Roots of $Ax^2 + Bx + C = 0$ $x_{1,2} = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$

$$x_{1,2} = \frac{4 \pm \sqrt{16 - 4 \cdot 2}}{4} = \frac{4 \pm \sqrt{8}}{4} = 1 \pm \frac{\sqrt{2}}{2}$$

$$2\left(x - \left(1 + \frac{\sqrt{2}}{2}\right)\right)\left(x - \left(1 - \frac{\sqrt{2}}{2}\right)\right) > 0$$

satisfied if both of them positive or negative

Positive $x > 1 + \frac{\sqrt{2}}{2}$ $x > 1 - \frac{\sqrt{2}}{2}$ Solution $x > 1 + \frac{\sqrt{2}}{2}$ $x \in (1 + \frac{\sqrt{2}}{2}, \infty)$

Negative $x < 1 + \frac{\sqrt{2}}{2}$ $x < 1 - \frac{\sqrt{2}}{2}$ Solution $x < 1 - \frac{\sqrt{2}}{2}$ $x \in (-\infty, 1 - \frac{\sqrt{2}}{2})$

Solution is the union of these sets $(-\infty, 1 - \frac{\sqrt{2}}{2}) \cup (1 + \frac{\sqrt{2}}{2}, \infty)$

$\textcircled{4}$ Solve the inequality $\frac{3}{x-1} < \frac{2}{x}$ and graph the solution set.

$$\frac{3}{x-1} < \frac{2}{x} \Rightarrow \frac{3}{x-1} - \frac{2}{x} < 0 \Rightarrow \frac{5x-2}{x(x-1)} < 0$$

$$x = 2/5 \text{ root}$$

$$x=0, x=1 \text{ undefined}$$

x	0	2/5	1
5x-2	-	0	+
x	-	0	+
x-1	-	-	0
5x-2	-	undef	+
x(x-1)	-	undef	+

Solution set

$$(-\infty, 0) \cup (2/5, 1)$$

The Absolute Value

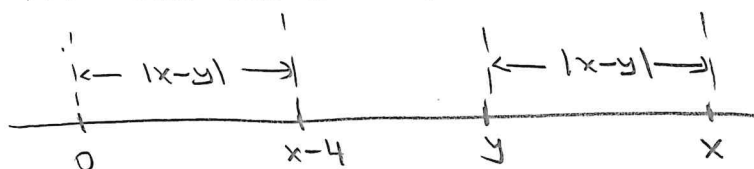
$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Absolute value bars.

$$|x| \geq 0 \text{ for every real number } x$$

$$|x| = 0 \text{ only if } x = 0$$

Absolute value represents the distance between points



$$|x-y| = \begin{cases} x-y & \text{if } x > y \\ y-x & \text{if } x < y \end{cases}$$

(b)

Properties of absolute values

① $| -a | = | a |$ ② $| ab | = | a | | b |$ and $\left| \frac{a}{b} \right| = \frac{| a |}{| b |}$

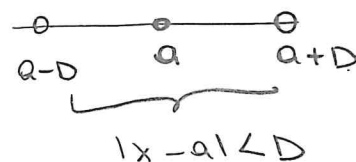
③ $| a \pm b | \leq | a | + | b |$ (the triangle inequality)

④ $| x - a | = D \Leftrightarrow \begin{matrix} x - a = -D \\ x = a - D \end{matrix} \text{ or } \begin{matrix} x - a = D \\ x = a + D \end{matrix}$

⑤ $| x - a | < D \Leftrightarrow a - D < x < a + D$

⑥ $| x - a | \leq D \Leftrightarrow a - D \leq x \leq a + D$

⑦ $| x - a | > D \Leftrightarrow \begin{matrix} x - a > D \\ x > D + a \end{matrix} \quad \begin{matrix} x - a < -D \\ x < a - D \end{matrix}$



a: center
D: radius

Ex 7 Solve ② $| 2x + 5 | \leq 3$ ⑤ $| 3x - 2 | \leq 1$ ③ $| x + 1 | = | x - 3 |$

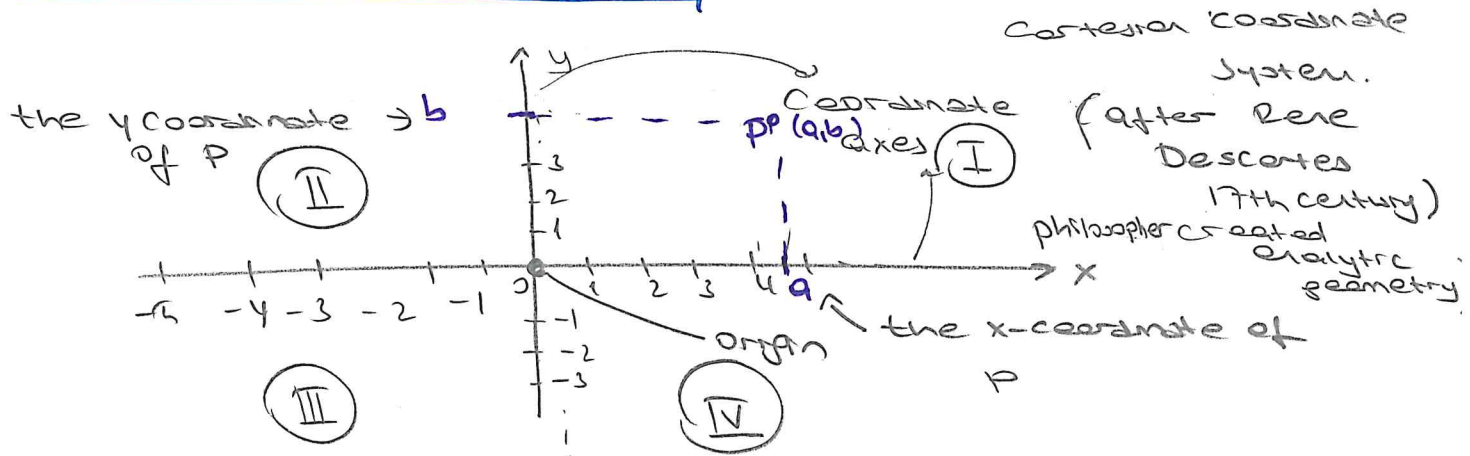
② $2x + 5 = 3 \Rightarrow x = -1$ ④ $| 5 - \frac{2}{x} | < 3 \Rightarrow 2x + 5 = -3 \Rightarrow x = -4$

⑤ $| 3x - 2 | \leq 1 \Rightarrow -1 \leq 3x - 2 \leq 1 \Rightarrow 1 \leq 3x \leq 3 \Rightarrow \frac{1}{3} \leq x \leq 1 \Rightarrow x \in [\frac{1}{3}, 1]$

③ $x + 1 = x - 3$ (No solution) or $x + 1 = -(x - 3) \Rightarrow 2x = 2 \Rightarrow x = 1$

④ $| 5 - \frac{2}{x} | < 3 \Rightarrow -3 < 5 - \frac{2}{x} < 3 \Rightarrow -8 < -\frac{2}{x} < -2 \Rightarrow 4 > \frac{1}{x} > 1 \Rightarrow \frac{1}{4} < x < 1$

Cartesian Coordinates in the Plane

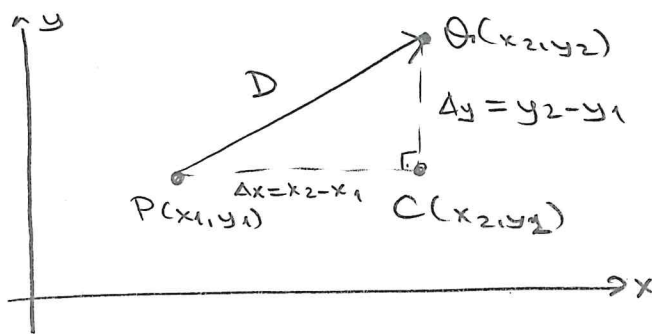


(I), (II), (III), (IV) are the four quadrants of Cartesian coordinate system

x: Horizontal axis

y: Vertical axis

Increments and Distances



Δx : x-increment

Δy : y-increment

The distance from P to Q is $D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

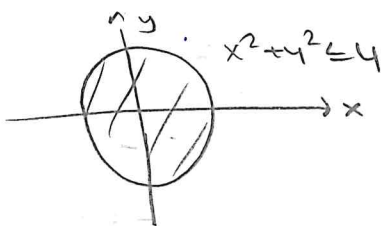
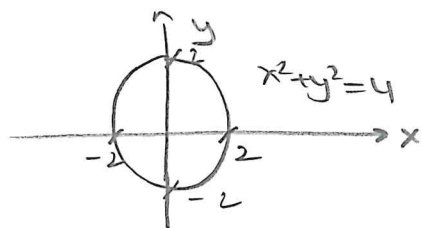
Ex 7 Find the movements in the coordinates of a particle that moves from A(3, -3) to B(-1, 2)

$$\Delta x = -1 - 3 = -4 \quad \Delta y = 2 - (-3) = 5$$

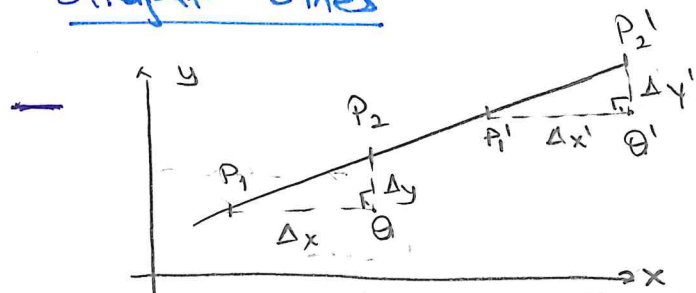
Find distance between A and B $|AB| = \sqrt{(-1-3)^2 + (2-(-3))^2} = \sqrt{41}$

GRAPHS

→ The graph of an equation (or inequality) involving the variables x and y is the set of all points $P(x,y)$ whose coordinate satisfy the equation (or inequality)

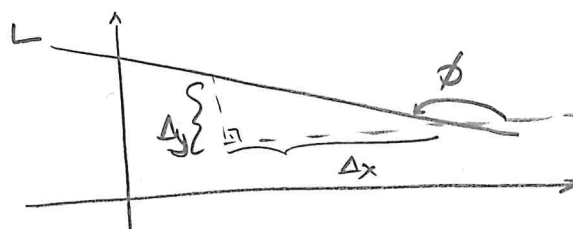
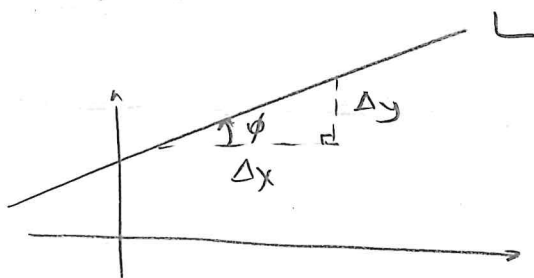
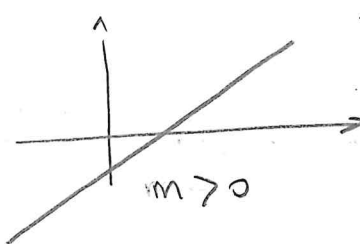


Straight Lines



$$m = \frac{\Delta y}{\Delta x} = \frac{\Delta y'}{\Delta x'}$$

Slope

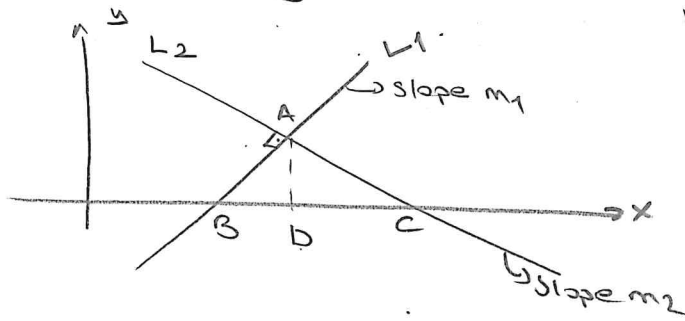


The line L has inclination ϕ (counterclockwise angle measured $0 \leq \phi \leq 180^\circ$ from positive direction of the x axis to the line)

$$m = \frac{\Delta y}{\Delta x} = \tan \phi$$

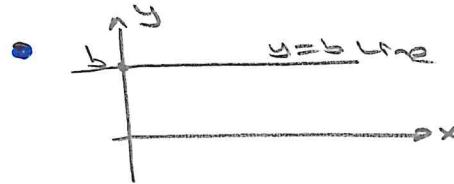
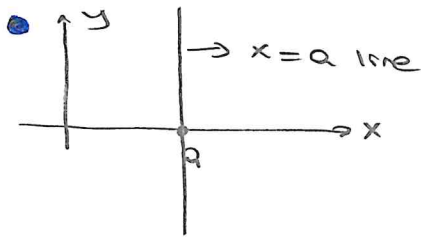
x axis $\phi = 0^\circ$, y axis $\phi = 90^\circ$

- If two nonvertical lines L_1 and L_2 , are perpendicular, their slopes m_1 and m_2 satisfy $m_1 \cdot m_2 = -1$.



$$m_1 = -\frac{1}{m_2} \quad \text{and} \quad m_2 = -\frac{1}{m_1}$$

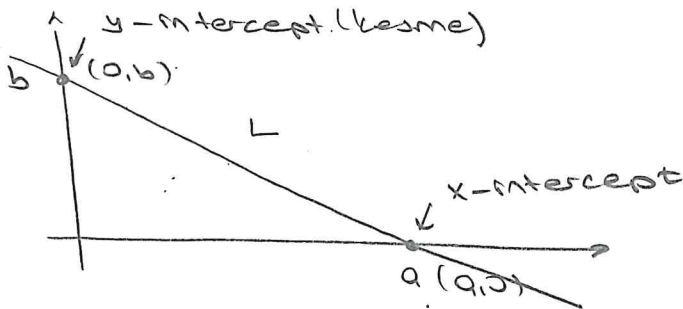
Equation of Lines



• The equation

$$y = m(x - x_1) + y_1 \quad \text{is the point-slope equation}$$

of the line that passes through the point (x_1, y_1) and has slope m .



$$y = mx + b$$

the slope-y-intercept equation

$$y = m(x - a)$$

the slope-x-intercept equation

The equation $Ax + By = C$ (where A and B are not both zero) is called the general linear equation in x and y

Ex 7 a) Find an equation of the line of slope -2 through the point $(1, 4)$

$$y - 4 = -2(x - 1) \quad y = -2x + 6$$

b) Find an equation of the line through the points $(1, -1)$ and $(3, 5)$

$$m = \frac{\Delta y}{\Delta x} = \frac{5 - (-1)}{3 - 1} = 3$$

Use $(1, -1)$

$$y - (-1) = 3(x - 1)$$

$$y = 3x - 3 - 1 = 3x - 4$$

$$\underline{y = 3x - 4}$$

Use $(3, 5)$ $y - 5 = 3(x - 3)$ $y - 5 = 3x - 9$ $\underline{y = 3x - 4}$

Graphs of Quadratic Equations

Circles and Disks

Circle

Standard equation of a circle

- The circle with center (h, k) and radius $a > 0$ has equation.

$$\underline{(x - h)^2 + (y - k)^2 = a^2}$$

- In particular, the circle with center at the origin $(0, 0)$ and radius a

$$\underline{x^2 + y^2 = a^2}$$

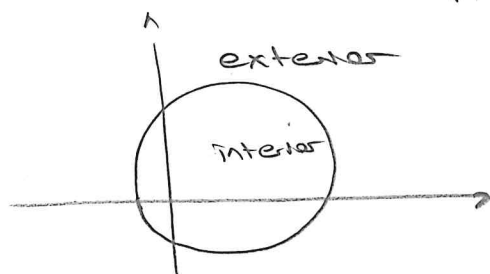
$$(x - h)^2 + (y - k)^2 = a^2 \Rightarrow x^2 - 2hx + h^2 + y^2 - 2ky + k^2 = a^2 \text{ write in form}$$

$$x^2 + y^2 + 2ax + 2by = c \rightarrow \text{Circle, when } c + a^2 + b^2 > 0$$

single point when $c + a^2 + b^2 = 0$

no points when $c + a^2 + b^2 < 0$

Disk



A disk of radius a with center (h, k)

$$(x - h)^2 + (y - k)^2 \leq a^2$$

Ex Find the center and radius of the circle $x^2 + y^2 - 4x + by = 3$

$$x^2 - 4x + y^2 + by = 3$$

$$(x-2)^2 + (y+3)^2 = 3 + 4 + 9$$

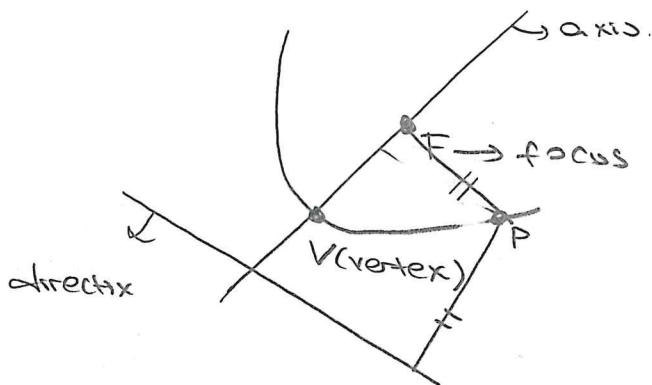
$$(x-2)^2 + (y+3)^2 = 16 \quad \text{center } (2, -3) \quad \text{radius} = 4$$

Parabolas

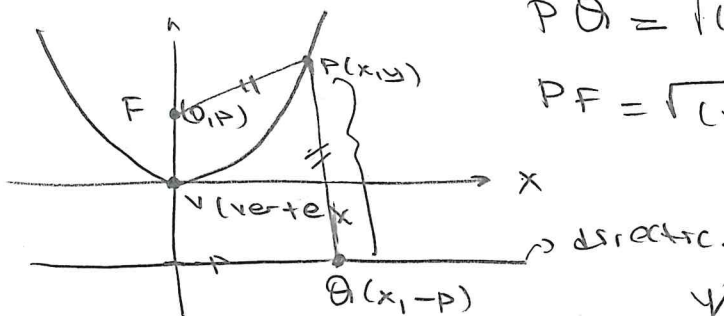
→ A parabola is a plane curve whose points are equidistant from a fixed point F (focus) and a fixed straight line L (directrix) that doesn't pass through F.

→ The line through F perpendicular to L is the parabola's axis

→ The point V where the axis meet the parabola is parabola's vertex



Ex Find an equation of the parabola having the point $F(0, p)$ as focus and the line L with equation $y = -p$ as directrix



$$PQ = \sqrt{(x-x)^2 + (y-(-p))^2} = \sqrt{y^2 + 2py + p^2}$$

$$PF = \sqrt{(x-0)^2 + (y-p)^2} = \sqrt{x^2 + y^2 - 2py + p^2}$$

$$PQ = PF$$

$$y^2 + 2py + p^2 = x^2 + y^2 - 2py + p^2$$

$$x^2 = 4py$$

$$y = \frac{x^2}{4p}$$

SCALING GRAPH

SHIFTING GRAPH

Shifting a graph HORIZONTALLY

$c > 0$, $x \leftrightarrow x - c$ Shifting c units to the right

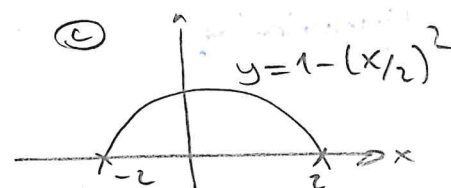
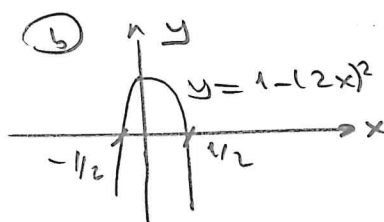
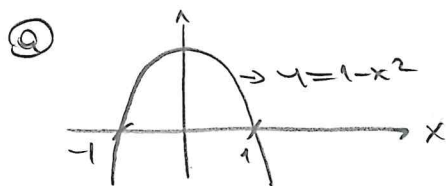
$c < 0$, $x \leftrightarrow x - c$ Shifting c units to the left

Shifting a graph VERTICALLY

$c > 0$, $y \leftrightarrow y - c$ Shifting c units to the upward.

$c < 0$, $y \leftrightarrow y - c$ Shifting c units to the downward

SCALING GRAPH



Ex Describe the graph $y = x^2 - 4x + 3$

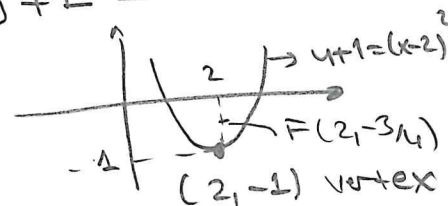
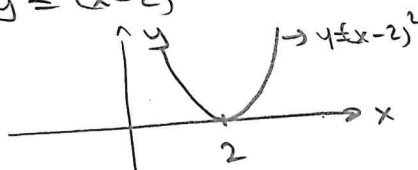
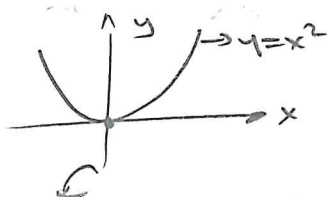
$$y = x^2 - 4x + 3 = (x - 2)^2 - 1$$

$y = x^2$ shift right 2

$$y = (x - 2)^2$$

shift downward 1

$$y + 1 = (x - 2)^2$$



From our previous

example on page 12

its focus is

$$y = \frac{x^2}{4p} \quad 4p = 1 \quad p = 1/4$$

$F(0, 1/4) \rightarrow$ focus

$y = -1/4 \rightarrow$ directrix

Shift your focus

$$\begin{aligned} &F(2, 1/4) \text{ 2 right} \\ &F(2, -3/4) \text{ 1 down} \end{aligned}$$

Vertex is the middle
between focus and
directrix $y = -5/4$

NOT $(x - h)^2 = 4p(y - k)$

$\neq (h, k + p) \rightarrow$ focus

$V(h, k) \rightarrow$ vertex

$y = k + p \rightarrow$ directrix

$$(x - 2)^2 = (y + 1)$$

$$p = 1/4 \quad h = 2 \quad k = -1$$

$$F(2, -1 + 1/4) = F(2, -3/4)$$

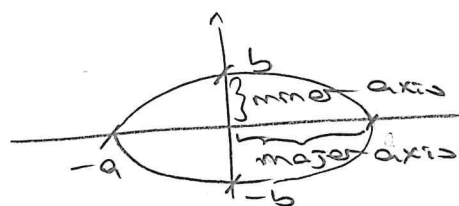
$$V(2, -1) \quad y = -1 - 1/4 = -5/4$$

Ellipses and Hyperbolas

Ellipse If a and b are positive numbers, the equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ represents a curve called an ellipse.}$$

that lies wholly within the rectangle $-a \leq x \leq a$ $-b \leq y \leq b$

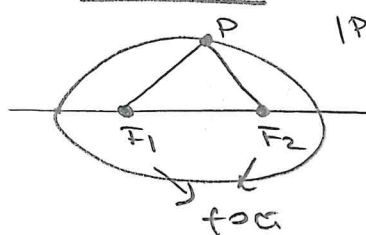


Principal axes The line segments from $(-a, 0)$ to $(a, 0)$ and from $(0, -b)$ to $(0, b)$.

The longer of two is the major axis

The shorter of two is the minor axis

General definition of ellipse



$$|PF_1| + |PF_2| = \text{const}$$

Hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

